

DS 8007 Advanced Data Visualization Project Proposal

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Project Definition: Bicycle thefts are a serious issue in Toronto. According to the Toronto Police Service Public Safety Data Portal, more than 1,900 [1] bicycles have been stolen in Toronto year-to-date. For policymakers to address this issue, they would like to know where, when, and the circumstances for why bicycles are being stolen. Our project aims to tackle this issue and these underlying questions by visualizing and analyzing bicycle theft patterns using open data from the Toronto Police Service. Insights gained from finding high-risk neighborhoods, seasonal and time-based trends, and relationships between characteristics such as bike type, cost, and theft status, can inform citizens to take preventative actions and policymakers to set appropriate rules and regulations to reduce and prevent bike theft occurrences.

Research Question: The key question this project aims to answer is: What trends and patterns can be found in bicycle thefts across Toronto in terms of where, when, and how they occur? It examines which geographical locations experience the most thefts; which days, months, and hours show higher activity; and how bike features such as make, model, color, cost, type, and speed relate to theft risk. The project also looks at environmental factors as they relate to theft status such as the type of premises where thefts happen and reporting behaviour, including the time gap between when a theft occurred and when it was reported.

Motivation: In Toronto, where summers are short and outdoor activity is limited, bicycle theft has become a major problem affecting both community safety and sustainable transportation. It discourages people from cycling, limiting eco-friendly travel and outdoor recreation. Understanding theft patterns can help design better municipal infrastructure, awareness campaigns, and safety measures and regulations. Solving this issue would encourage families and individuals to ride confidently and more often, supporting a healthier and greener city. [2]

Dataset: The dataset used in this project is the Toronto Police Service – Bicycle Thefts Open Data, covering reported bicycle thefts in Toronto from 2013 to 2025. It contains around 32,000 records in CSV format.

Key attributes include ID fields (EVENT_UNIQUE_ID, PRIMARY_OFFENSE), temporal features (OCC_DATE, OCC_YEAR, OCC_MONTH, OCC_DOW, OCC_DAY), spatial features (DIVISION, NEIGHBOURHOOD, LAT, LONG), bicycle details (MAKE, MODEL, TYPE, SPEED, COLOR, COST), contextual features (LOCATION_TYPE, PREMISES_TYPE), and outcome (STATUS: Stolen / Recovered / Unknown).

Dataset: <https://data.torontopolice.on.ca/datasets/TorontoPS::bicycle-thefts-open-data/about>

References:

[1] City of Toronto, "Bicycle Theft," Toronto Police Service Open Data, [Online]. Available:

<https://data.torontopolice.on.ca/pages/bicycle-thefts>.

[2] T. S. Ripa, T. F. H. Khan, M. Rokunuzzaman, M. R. Rawshan, S. A. Oishi, and K. M. Zim, "Bicycle Theft Detection by using YOLOv5s and DeepSORT algorithm," in *Proc. 2024 Int. Conf. Recent Progresses in Science, Engineering and Technology (ICRPSET)*, Rajshahi, Bangladesh, 2024, pp. 1–5, doi: 10.1109/ICRPSET64863.2024.10955901.